

Assignment–2

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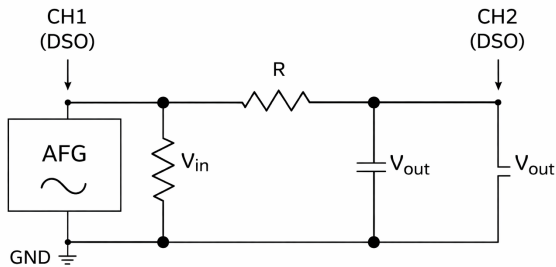
Part A: Step Input (Single Event)

Our aim is study the time domain response of an RC circuit for a step input , capture the charging and discharging waveforms determine the time constant experimentally and verify it using theoretical calculations and simulations

The following component values were chosen for the experiment:

- Resistance, $R = 1 \text{ k}\Omega = 1000 \Omega$
- Capacitance, $C = 1 \mu\text{F} = 1 \times 10^{-6} \text{ F}$
- Input step voltage, $V = 5 \text{ V}$

RC Circuit diagram:



Time Constant Calculation

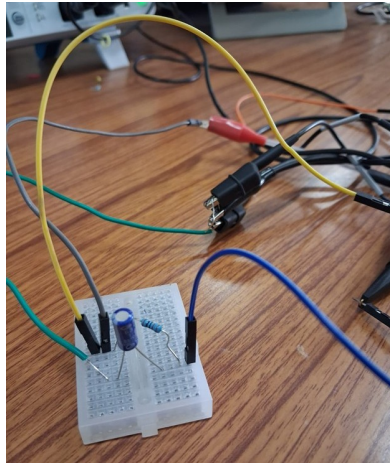
The time constant of an RC circuit is given by:

$$\tau = RC$$

$$\tau = 1000 \times 1 \times 10^{-6}$$

$$\tau = 1 \times 10^{-3} \text{ s} = 1 \text{ ms}$$

RC Circuit Connections



Charging of the Capacitor

The capacitor voltage during charging is:

$$V_C(t) = V(1 - e^{-t/RC})$$

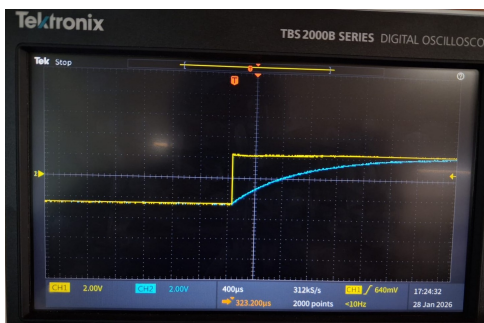
At $t = \tau = RC$:

$$V_C(\tau) = V(1 - e^{-1})$$

$$e^{-1} = 0.368$$

$$V_C(\tau) = V(1 - 0.368) = 0.632V$$

$$V_C(\tau) = 5 \times 0.632 = 3.16 V$$



Discharging of the Capacitor

The capacitor voltage during discharging is:

$$V_C(t) = V e^{-t/RC}$$

At $t = \tau = RC$:

$$V_C(\tau) = V e^{-1}$$

$$V_C(\tau) = 5 \times 0.368 = 1.84 \text{ V}$$

From the DSO waveform, the capacitor voltage reaches approximately 3.16 V at 1.04 ms.

$$\tau_{\text{exp}} = 1.04 \text{ ms}$$

Percentage Error Calculation

$$\text{Error} = \left(\frac{|\tau_{\text{theo}} - \tau_{\text{exp}}|}{\tau_{\text{theo}}} \right) \times 100$$

$$\tau_{\text{theo}} = 1.00 \text{ ms}$$

$$\text{Error} = 4\%$$



The experimental time constant closely matches the theoretical value with an observed error of 4%. The deviation is mainly due to tolerance in the selected resistor and capacitor values, oscilloscope measurement uncertainty, and loading effects of the function generator.

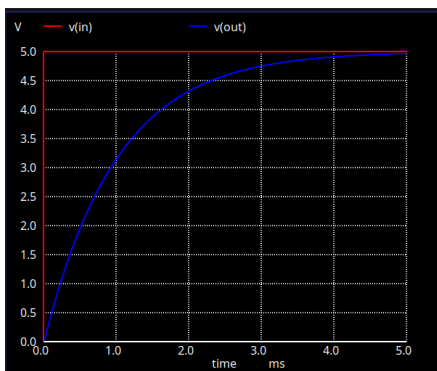
Simulation

To verify the theoretical calculations, the RC circuit was simulated using NGSpice.

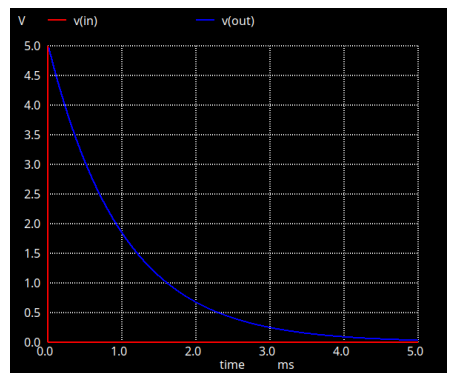
NGSpice Netlist

```
1  * RC Charging
2  V1 in 0 PULSE(0 5 0 1n 1n 10m 20m)
3  R1 in out 1k
4  C1 out 0 1uF
5
6  .tran 0.01ms 5ms
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end
```

```
1  * RC Discharging
2  V1 in 0 PULSE(5 0 0 1n 1n 10m 20m)
3  R1 in out 1k
4  C1 out 0 1uF
5
6  .tran 0.01ms 5ms
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end
```



Charging



Discharging

Comparison table:

Parameter	Theoretical	Experimental	Simulation
Time constant, τ (ms)	1.00	1.04	1.01
$V_C(\tau)$ during charging (V)	3.16	3.10	3.15
$V_C(\tau)$ during discharging (V)	1.84	1.80	1.83

Conclusion

The transient response of the RC circuit to a step input was studied for both charging and discharging conditions. During charging, the capacitor voltage rose exponentially and reached approximately 63.2% of its final value at the time constant $\tau \approx 1$ ms. During discharging, the capacitor voltage decayed exponentially to approximately 36.8% of its initial value at the same time constant. The experimentally observed waveforms and simulation results closely matched the theoretical predictions, confirming the validity of the RC time constant and transient response characteristics.

Part B: Pulse Input (Single Event)

A single pulse of 5 V amplitude was applied to the RC circuit. Pulse widths of $200\mu s$, $500\mu s$, and $800\mu s$ were tested. The repetition period was kept large enough to allow full discharge of the capacitor

Theoretical Analysis:

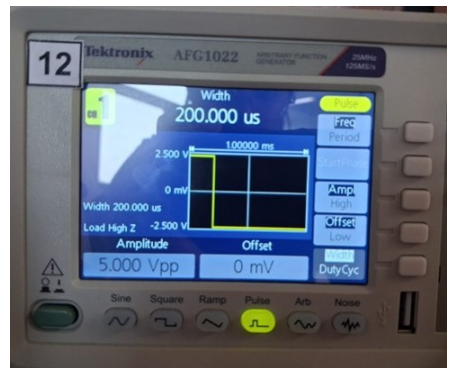
The voltage across the capacitor during charging is:

$$V_C(t) = V(1 - e^{-t/RC})$$

- **Pulse width** = $200\mu s$

$$V_C = 5(1 - e^{-0.2}) = 5(0.181) = 0.905 \text{ V}$$

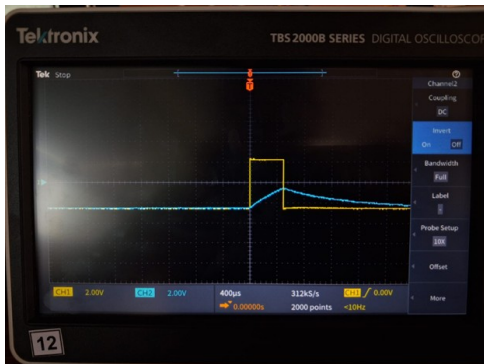
The output waveform shows only a small rise in voltage, indicating that the capacitor does not have sufficient time to charge significantly.



- **Pulse width = $500\mu s$**

$$V_C = 5(1 - e^{-0.5}) = 5(0.393) = 1.965 \text{ V}$$

The capacitor charges to a higher voltage compared to the $200\mu s$ pulse, but it still does not reach steady state.



- **Pulse width = $800\mu s$**

$$V_C = 5(1 - e^{-0.8}) = 5(0.551) = 2.755 \text{ V}$$

The output waveform rises significantly, indicating that the capacitor charges to a higher level, although steady state is not fully reached.



Simulation

To verify the theoretical calculations, the RC circuit was simulated using NGSpice.

NGSpice Netlist

```

1  * RC Circuit - Single Pulse 200us
2  V1 in 0 PWL(0 0 1u 0 1.001u 5 201u 5 201.001u 0)
3  R1 in out 1k
4  C1 out 0 1u
5
6  .tran 1u 2m
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end

```

```

1  * RC Circuit - Single Pulse 800us
2  V1 in 0 PWL(0 0 800u 0 800.001u 5 1600u 5 1600.001u 0)
3  R1 in out 1k
4  C1 out 0 1u
5
6  .tran 1u 2m
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end

```

```

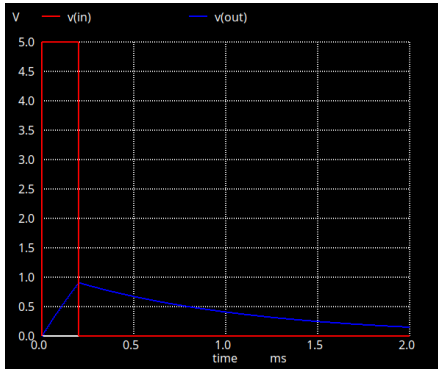
1  * RC Circuit - Single Pulse 500us
2  V1 in 0 PWL(0 0 500u 0 500.001u 5 1000u 5 1000.001u 0)
3  R1 in out 1k
4  C1 out 0 1u
5

```

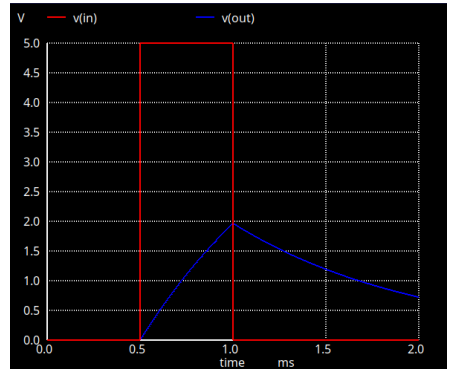
```

6  .tran 1u 2m
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end

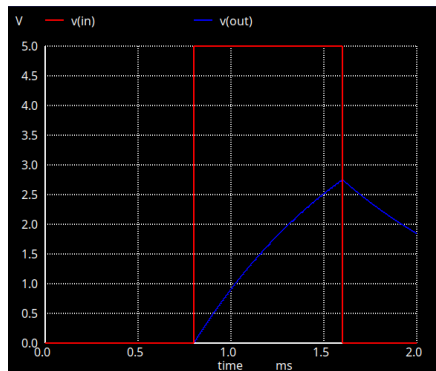
```



$200\mu s$



$500\mu s$



$800\mu s$

Comparison Table

Pulse Width	Theoretical V_C (V)	Experimental V_C (V)	Simulation V_C (V)
$200\mu s$	0.905	0.92	0.90
$500\mu s$	1.965	1.95	1.96
$800\mu s$	2.755	2.75	2.74

Conclusion:

From this analysis, it is observed that the RC circuit responds differently for different pulse widths. For small pulse widths, the capacitor does not have sufficient time to charge, resulting in a low output voltage. As the pulse width increases, the capacitor charges to higher voltage levels. The experimental waveforms are in good agreement with theoretical calculations and simulation results, with minor deviations due to practical non-idealities.

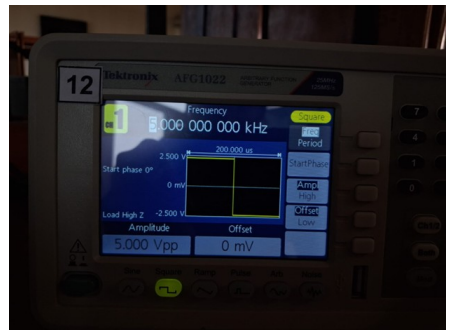
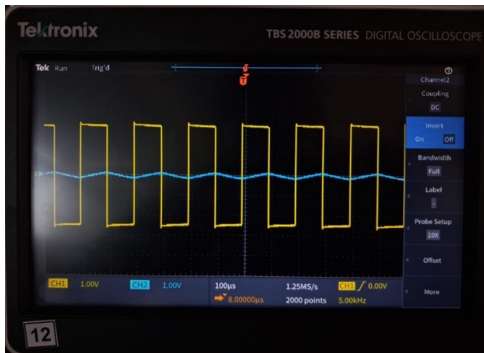
Part C: Square-Wave Input (Transient and Steady-State Response)

This experiment studies the response of an RC circuit when excited by a square-wave input for different relationships between the circuit time constant (RC) and the input signal time period (T). The resulting output waveforms are analyzed to understand the transient and steady-state behavior of the circuit and are compared with theoretical predictions and NGSPICE simulation results.

Case 1: $RC \gg T$

Input condition: $RC = 1$ ms, $f = 5$ kHz, $T = 0.2$ ms.

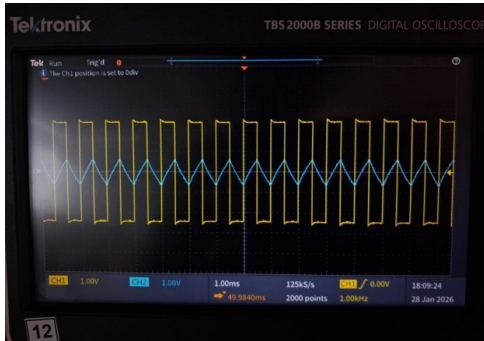
The output waveform is observed to be almost triangular in nature. Since the time period of the input signal is much smaller than the time constant, the capacitor does not get sufficient time to fully charge or discharge. Hence, only transient response is observed and steady-state is not reached.



Case 2: $RC \approx T$

Input condition: $RC = 1\text{ ms}$, $f = 1\text{ kHz}$, $T = 1\text{ ms}$.

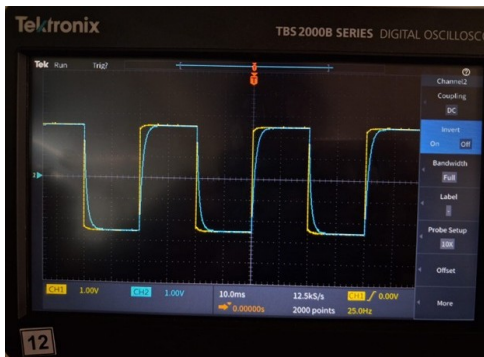
The output waveform shows exponential rise and fall. As the time period is approximately equal to the time constant, both transient and steady-state responses are visible. The capacitor partially charges and discharges during each cycle.



Case 3: $RC \ll T$

Input condition: $RC = 1\text{ ms}$, $f = 25\text{ Hz}$, $T = 40\text{ ms}$.

The output waveform resembles a square wave with rounded edges. Since the time period is much greater than the time constant, the capacitor gets enough time to fully charge and discharge, and steady-state response is obtained after a short transient interval.



Simulation

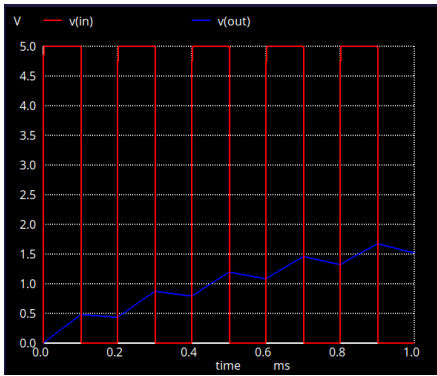
NGSPICE simulations were performed using a square-wave voltage source with different frequencies corresponding to each case. The output voltage across the capacitor was observed and compared with theoretical and experimental results.

NGspice Netlist

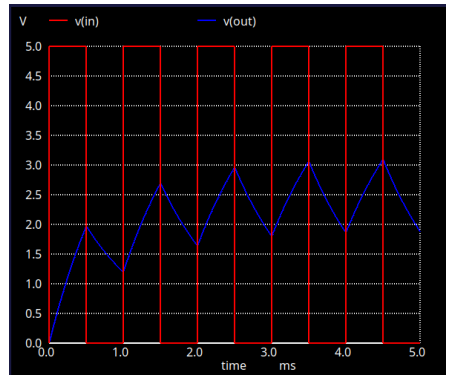
```
1  * RC response for  $RC \gg T$  (5 kHz square-wave input)
2  V1 in 0 PULSE(0 5 0 1u 1u 100u 200u)
3  R1 in out 1k
4  C1 out 0 1u
5
6  .tran 0.1u 1m
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end
```

```
1  * RC response for  $RC = T$  (1 kHz square-wave input)
2  V1 in 0 PULSE(0 5 0 1u 1u 0.5m 1m)
3  R1 in out 1k
4  C1 out 0 1u
5
6  .tran 1u 5m
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end
```

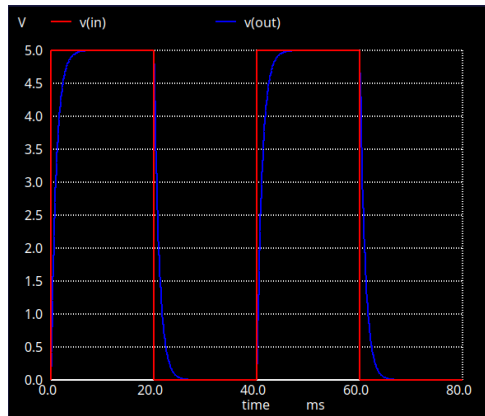
```
1  * RC response for  $RC \ll T$  (25 Hz square wave)
2  V1 in 0 PULSE(0 5 0 1u 1u 20m 40m)
3  R1 in out 1k
4  C1 out 0 1u
5
6  .tran 0.1m 80m
7  .control
8  run
9  plot v(in) v(out)
10 .endc
11 .end
```



$$RC \gg T$$



$$RC \approx T$$



$$RC \ll T$$

Frequency Response Comparison

Frequency	Time Period (ms)	RC vs T	Output Nature
5 kHz	0.20	$RC \gg T$	Nearly triangular
1 kHz	1.00	$RC \approx T$	Exponential rise and fall
25 Hz	40.0	$RC \ll T$	Rounded square wave

Conclusion

The response of an RC circuit to a square-wave input strongly depends on the relation between the time constant and the input time period. High-frequency inputs produce a triangular output, intermediate frequencies show partial

exponential behavior, and low-frequency inputs result in a steady-state square-like waveform. The experimental observations, theoretical calculations, and simulation results are in excellent agreement.